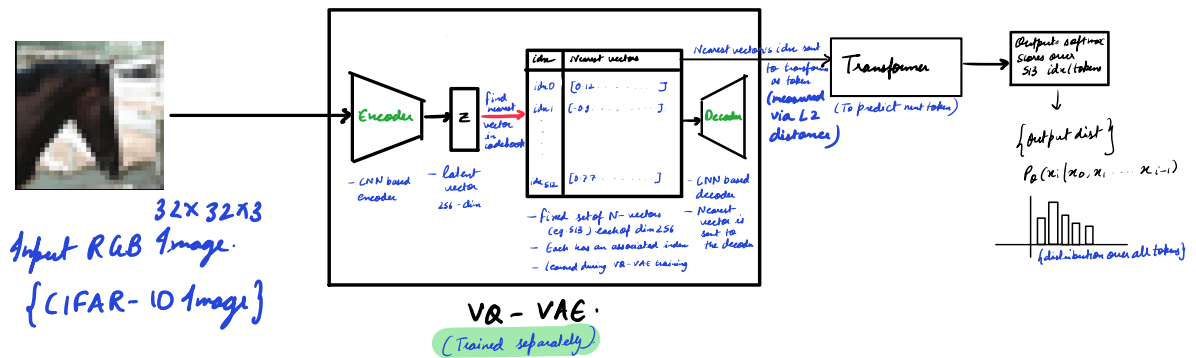


Autoregressive models for Image generation

Instead of generating the entire image at once like the diffusion architecture, an autoregressive model generates the image one token at a time similar to how transformers generate text.

full Pipeline



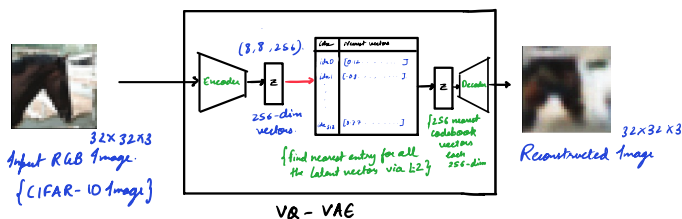
Why VQ-VAE?

Transformers need finite discrete, tokens not continuous vectors.

The codebook creates a fixed vocab of 513 tokens for the image and then transformer predicts which of the 513 idx / tokens come next.

Training

① Phase 1. Train VQ-VAE:



Losses utilized for training:

1. Reconstruction Loss: Comparing reconstructed image with original image.

$$\| \text{original} - \text{image} - \text{reconstructed} \|_2$$
2. Codebook Loss: Push codebook entries closer to the encoder outputs.

$$\| z - c \|_2$$

Stop gradient, so that encoder weights don't update from this loss.
 $z \rightarrow$ raw encoder outputs.
 $c \rightarrow$ nearest codebook entry
3. Commitment Loss: Push encoder outputs to stay close to their nearest codebook entry.

$$\| z - c - \text{sg}(c) \|_2$$

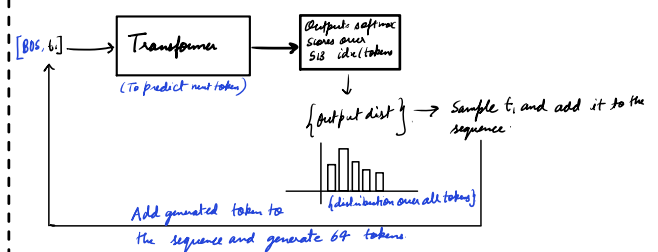
{only updates encoder weights}

Total Loss = reconstruction + codebook + β * commitment
 (β is typically 0.25)

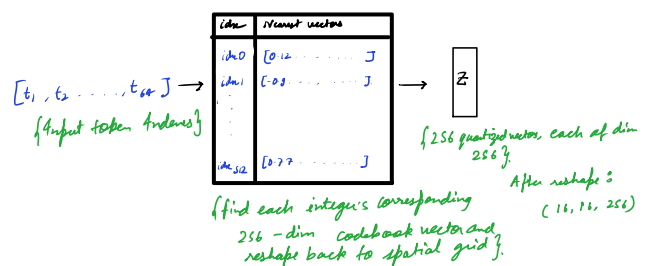
Inference

Inference is performed over 3 steps:

Step 1: Transformer samples 64 token autoregressively:

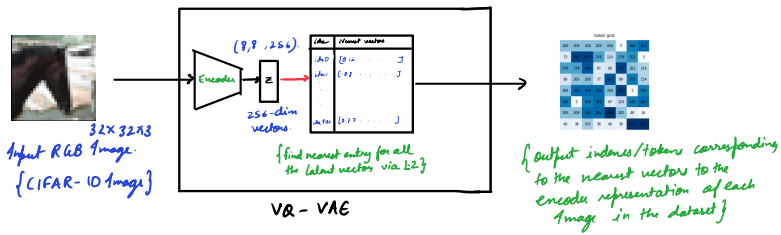


Step 2: Codebook lookup:

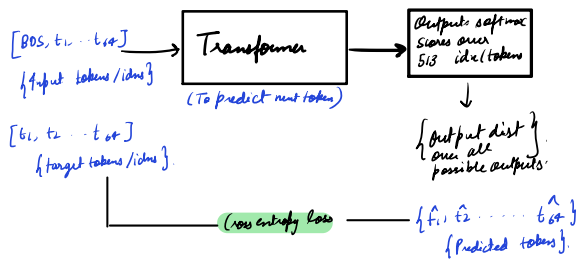


② Phase 2 : Train the Transformer

1. Build the dataset : Run every image through trained VR-VAE (frozen)
 & save resulting 64-dim integer sequences



2. Train the transformer on the prepared dataset :



Step 3 : VR-VAE Decoder :

